

Chapter 5

Lorraine Gaudio

Version April 21, 2026



BOISE STATE UNIVERSITY

Contents

1. Overview

In this chapter, you'll learn how to turn your R Notebook into a shareable HTML report that another person can read **without opening RStudio!** You'll start by using Markdown to make your notebook readable and navigable by adding a date in the YAML header and headings that create an outline so your work looks like a structured report. Then you'll generate an HTML version of the notebook and learn how to recognize (and troubleshoot) common problems when the report fails to build. Because modern researchers often use AI tools during debugging, you'll also learn how to write well-structured Markdown prompts to ask an LLM for targeted help without delegating your thinking. This matters now because, moving forward, you will be evaluated on whether your workflow is understandable and reproducible (CLO1), whether you can diagnose problems when something breaks (CLO2), and whether you can document responsible help-seeking and tool use as part of your process (CLO3).

In Chapter Five you will learn how to:

- #□ Format your R Notebook using Markdown
- □ Structure prompts for Large Language Models (LLMs)

2. R Notebook -> HTML Report

□ Please open a new R Notebook and save it in your course folder named chapter5_notes.Rmd. Use this document to take notes and practice the chapter content.

1. Open RStudio → File → New File → R Notebook.
2. Save it in your course folder as: chapter5_notes.Rmd.
3. In your YAML header at the top, make sure the title matches this chapter.

2.1 YAML Date Field

The YAML header is the metadata “label” for your notebook. It controls document settings like title and output format. A basic YAML header looks like this:

```
---
title: "Untitled"
output: html_notebook
---
```

To add a date to your notebook, edit the YAML header to include a date: field.

Method A. **Hard Code**

Manually type the date you created the notebook. Include quotation marks around the date.

```
---
title: "Chapter 5 Practice"
date: "January 13, 2026"
output: html_notebook
---
```

Method B. **Dynamic Date**

Use R code to generate the current date automatically each time you knit the document. The inline R expression ‘`r Sys.Date()`’ retrieves the current date from your computer’s operating system. This follows the syntax: backtick (`), then r, a space, Sys.Date(), and a closing backtick (`). The backtick is located above the Tab key on most keyboards.

```
---
title: "Chapter 5 Practice"
date: "`r Sys.Date()`"
output: html_notebook
---
```